Notes 01.07: Mastering Visual Studio Code for AI/ML Engineering

1. Why Visual Studio Code is the Premier AI Editor

Microsoft Visual Studio Code (VS Code) has become the gold-standard code editor in data science and AI engineering. It seamlessly combines:

- A lightning-fast, lightweight text editor.
- Built-in interactive **Jupyter Notebook (`.ipynb`)** execution engine.
- Integrated multi-shell terminal (PowerShell, Bash, WSL).
- Rich debugging with variable inspection and memory inspection.
- Native Git source control integration.

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2. Essential Extensions for AI Engineers

Search and install from the VS Code Extensions Marketplace (`Ctrl+Shift+X`):

1. **Python (by Microsoft):** Provides IntelliSense, linting, code formatting, and debugging.
2. **Pylance:** Fast, feature-rich language support with type checking.
3. **Jupyter (by Microsoft):** Enables running interactive notebook cells directly inside VS Code without launching a separate browser server.

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3. Selecting the Correct Python Interpreter

When working inside VS Code with virtual environments, you must tell the editor which Python binary to use:

1. Press `Ctrl + Shift + P` (or `Cmd + Shift + P` on Mac) to open the Command Palette.
2. Type and select: `Python: Select Interpreter` .
3. Choose the interpreter located inside your project's `.venv` or `aiml_env` directory (e.g., `./aiml_env/Scripts/python.exe`).
4. The selected environment will now display in the bottom-right status bar.

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4. Script (`.py`) vs Notebook (`.ipynb`) Workflow

Feature	Python Script (`.py`)	Jupyter Notebook (`.ipynb`)
Execution Model	Entire file executes top-to-bottom	Interactive cell-by-cell execution
State Persistence	Memory cleared when process exits	Variables, models, and data remain in RAM
Best Used For	Production pipelines, microservices, classes	Exploratory Data Analysis, model tuning, plotting